MILITARY SPECIFICATION

CLOTH; COTTON, TWILL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for one type of cotton twill cloth.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-P-1134 - Packaging and Packing of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics)

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 8305

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LAWS AND REGULATIONS

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained from the Federal Trade Commission, Washington, DC 20580.)

3. REQUIREMENTS

3.1 Material. The cloth shall be made from cotton of a grade and staple length that will produce a material meeting the requirements specified herein.

3.2 Color. Unless otherwise specified the color shall be Olive Green 107.

3.2.1 Colorfastness. The dyed cloth shall show "good" fastness to Cold water, light, perspiration, laundering, dry cleaning and crocking when tested as specified in 4.5.

3.3 Physical requirements. The finished cloth shall conform to the physical requirements shown in table I when tested as specified in 4.5.

TABLE I. Physical requirements

Weight oz./sq. yd.		Yarns per inch (Min.)		Breaking strength (Min.) pounds		Tearing strength (Min.) pounds		Air permeability cu. ft./min. sq. ft. (Max.)
Min.	Max.	Warp	Filling	Warp	Filling	Warp	Filling	
5.6	6.25	190	92	145	60	3.0	3.0	13

3.3.1 Weave. The weave shall be 2 up and 2 down twill.

3.3.2 Width. The width shall be not less than 36 inches nor more than 37 inches.

3.4 Nonfibrous materials. The cloth shall not exceed 2.0 percent starch and protein content including chloroform-soluble and water-soluble material when tested as specified 4.5.

3.5 Dimensional stability. The finished cloth shall not shrink more than 1.5 percent in either the warp or filling direction when tested as specified in 4.5.

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3.6 Length and put up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards. Each length shall be put up in flat folded pieces or on full width rolls as specified in PPP-P-1134.

3.7 Fiber identification. Each roll or piece of cloth shall be labeled, ticketed, or invoiced for fiber content in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act.

3.8 Workmanship. Workmanship shall be in accordance with high-grade commercial practice covering manufacture of cloth of this quality. Fibers shall be thoroughly cleaned, combed to remove short fibers, and evenly spun into yarns of the proper size to produce the texture and weight required. Yarns shall be woven into fabric that shall be free from manufacturing imperfections, and the fabric uniformly dyed the desired shade.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection. Inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3 Examination of product. All material submitted for acceptance shall be inspected for conformance with material, workmanship, yarn, and weave requirements. In the event that visual examination indicates such compliance to be questionable, the procuring activity may require appropriate tests on selected specimens in accordance with FED-STD-191.

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4.3.1 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-1134 to determine that packaging, packing and marking comply with the section 5 requirements.

4.3.1.1 Examination for compliance with Textile Fiber Products Identification Act. An examination shall be made to insure that each unit (roll or fold) is labeled for fiber content in accordance with the Textile Fiber Products Identification Act.

4.4 Selection of representative sample. One sample, 1 yard in length and the full width of the cloth, shall be selected to represent each roll, bolt, or fold (approximately 500 yards) of cloth submitted for acceptance. Additional samples may be randomly selected for testing when considered necessary by the procuring activity.

4.5 Testing of the end item. The methods of testing specified in FED-STD-191, wherever applicable and as listed in table II shall be followed. The physical and chemical values specified in section 3, except where otherwise indicated, apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values utilized in expressing the final result. The sample unit shall be 2 yards, full width of the dyed cloth. The lot size shall be expressed in units of one yard. The sample size shall be as follows and the lot shall be unacceptable if one or more sample units fail to meet any requirements specified.

Lot size (yards)	Sample size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE II. Test methods

Characteristic	Requirement paragraph	Test method
Colorfastness:		
Cold water	3.2.1	5630
Light	3.2.1	5660
Perspiration	3.2.1	5680
Laundering	3.2.1	5610
Dry cleaning	3.2.1	5621
Crocking	3.2.1	5651
Weight	3.3	5041

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TABLE II. Test methods (cont'd)

Characteristic	Requirement paragraph	Test method
Yarns per inch:		
Warp	3.3	5050
Filling	3.3	5050
Breaking strength:		
Warp	3.3	5100
Filling	3.3	5100
Tearing strength:		
Warp	3.3	5134
Filling	3.3	5134
Air permeability	3.3	5452
Weave	3.3.1	Visual 1/
Nonfibrous materials	3.4	2611

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

5. PREPARATION FOR DELIVERY

5.1 Put-up and packaging. Put-up and packaging shall be Level A or C as specified (see 6.2).

5.1.1 Levels A and C. The cloth shall be put-up and packaged in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1134.

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6. NOTES

6.1 Intended use. The cloth covered by this specification is intended for use in the SPH-4 helment.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Color of cloth required if other than specified (see 3.2).
(c) Length of roll required if other than specified (see 3.6).
(d) Selection of applicable put-up, packaging, and packing (see 5.1 and 5.2).

6.3 This specification contains the basic requirements of cancelled Military Specification MIL-C-5039(USAF).

Custodian:

Army - GL

Preparing activity:

Army - GL

Project No. 8305-A496

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE
MIL-C-43921(GL) - CLOTH; COTTON, TWILL

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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